

Newsletter Subscription and Delivery System

Functional Requirements:

1. Create and Manage Newsletters with Subscriber Lists
 - Ability to add/delete news letters
 - Ability to add/remove subscribers for newsletters
2. Newsletters can schedule sending of emails to their subscribers at a specific date and time trigger. The system should efficiently deliver email newsletter to all subscriber emails. We will support two models:
 - Recurring model (try to send a email every 7 days for example)
 - Explicit send/push model (breaking news)
3. The system will support two basic templates when sending email, Plain Text and HTML.

Non-Functional Requirements:

1. The system should scalably handle a large number of newsletters and potentially large mailing lists.
2. The system should scalably be able to publish and deliver emails to ALL subscribers in a mailing list.
3. Reliability is very important, no subscriber should get lost, if a subscriber does not get a newsletter email that could be \$\$ for the company.
4. Latency is important. Some delay may be acceptable for recurring emails (+/- 30 minutes) but breaking news needs to have the lowest latency. This might be indicating some sort of priority system.

Assumptions Being Made:

1. Newsletters use our service for the scheduling and delivery of their email newsletters, but they themselves are responsible for providing us with the content. Our service is NOT a news feed.
2. We are not an email service, we will coordinate with other email API's to actually send the emails out to inboxes.

Plan of Attack on this Problem:

1. Define the Data Model and the DB Table Definition
2. Spec out what the Server Side API might look like (for the most important functional cases).
3. Think through the pressure points and thundering herd/failure points in the system. Think of how to handle them scalably.
4. Spec out a E2E diagram of the backend (with optional data flow)

